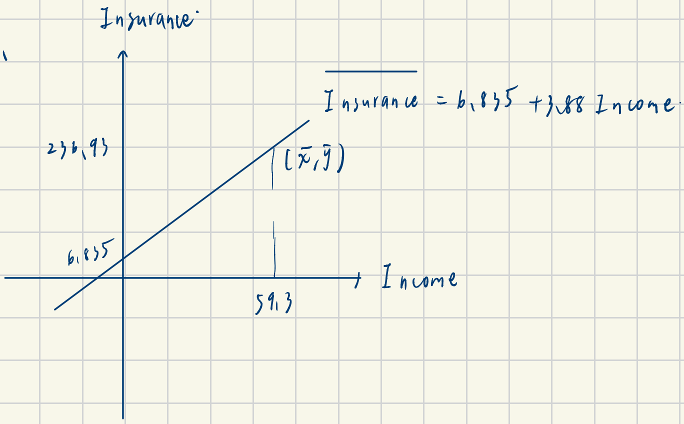
3.18

a.



b. β_2 的 95% CI 為 $b_2 \pm t_{0.025}(18) \times se(b_2)$

$= [\,3.88 - 2.101 \times 0.112\,,\ 3.88 + 2.101 \times 0.112\,]$

$= [\,3.645\,,\ 4.115\,]$ (千元)

我們有95%信心，家庭收入每增加1000元，該家庭的人壽保險金額會增加3.645到4.115元

(信心來自於可以重複抽樣)

c.

求 $E(Y|X=100)$ 之 99% C.I.

$\hat{Y}_0 = 6.885 + 3.88 \times 100 = 394.835$

$Var(\hat{Y}_0) = Var(b_1) + X_0^2 Var(b_2) + 2X_0 Cov(b_1, b_2)$

$= 7.383^2 + 100^2 \times 0.112^2 + 2 \times 100 \times (-0.746)$

$= 30.7487$

$se(\hat{Y}_0) = 5.545\ ,\quad t_{0.005}(18) = 2.878$

$E(Y|X=100)$ 之 99% C.I. 為

$[\,394.835 - 2.878 \times 5.545\,,\ 394.835 + 2.878 \times 5.545\,]$

$= [\,378.877\,,\ 410.813\,]$

d.

$\begin{cases} H_0: \beta_2 = 5 \\ H_a: \beta_2 \neq 5 \end{cases}$

$\alpha = 0.05$

test statistic: $t = \dfrac{b_2 - 5}{se(b_2)} \overset{H_0}{\sim} t(18)$

$RR = \{\, t \mid |t| > t_{0.025}(18) = 2.101 \,\}$

$t^* = \dfrac{3.88 - 5}{0.112} = -10$

$t^* \in RR$, reject H_0，有充分的證據證明 $\beta_2 \neq 5$

e、 $\begin{cases} H_0: \beta_2 \le 1 \\ H_a: \beta_2 > 1 \end{cases}$

$\alpha = 0.01$

test statistic $= t = \dfrac{b_2 - 1}{se(b_2)} \overset{H_0}{\sim} t(18)$

$RR = \{ t \mid t > t_{0.01}(18) = 2.552 \}$

$t^* = \dfrac{3.88 - 1}{0.112} = 25.71$

$t^* \in RR$, reject H_0, 有充足證據顯示價格增加幅度大於1.

3.23

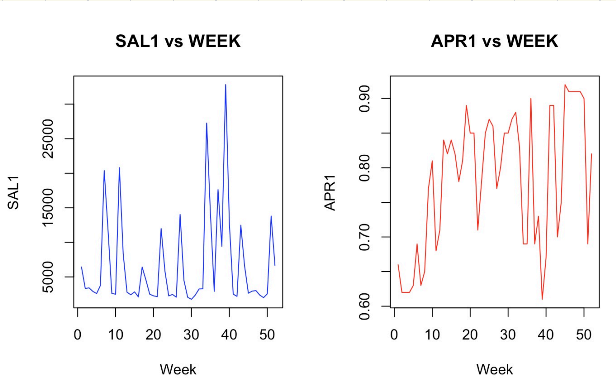
```
Call:
lm(formula = price ~ I(sqft^2), data = collegetown)

Residuals:
    Min      1Q  Median      3Q     Max 
-383.67  -48.39   -7.50   38.75  469.70 

Coefficients:
             Estimate Std. Error t value Pr(>|t|)    
(Intercept) 93.565854   6.072226   15.41   <2e-16 ***
I(sqft^2)    0.184519   0.005256   35.11   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 92.08 on 498 degrees of freedom
Multiple R-squared:  0.7122,    Adjusted R-squared:  0.7117 
F-statistic:  1233 on 1 and 498 DF,  p-value: < 2.2e-16
```

```
========== Part (a) ==========
H0: 40*a2 <= 13  vs  H1: 40*a2 > 13 (右尾, 5%)

邊際效果 (40 * a2_hat)  = 7.3808 千美元 = 7381 美元
t 統計量                = -26.7288
臨界值 (右尾 5%)         = 1.6479
p-value                 = 1
結論: t = -26.73 < 臨界值 1.65 → 無法拒絕 H0

========== Part (b) ==========
H0: 80*a2 <= 13  vs  H1: 80*a2 > 13 (右尾, 5%)

邊際效果 (80 * a2_hat)  = 14.7615 千美元 = 14762 美元
t 統計量                = 4.1895
臨界值 (右尾 5%)         = 1.6479
p-value                 = 1.7e-05
結論: t = 4.19 > 臨界值 1.65 → 拒絕 H0

========== Part (c) ==========
預測平均售價 = 167.3735 千美元 = $ 167373
95% CI = [ 158.0481 , 176.6988 ] 千美元

========== Part (d) ==========
sqft = 20 的房子數量    = 3 棟
樣本實際平均售價         = 163.3333 千美元 = $ 163333
95% CI                  = [$ 158048 , $ 176699 ]
樣本均值在CI內？          → 是，與預測相容
```

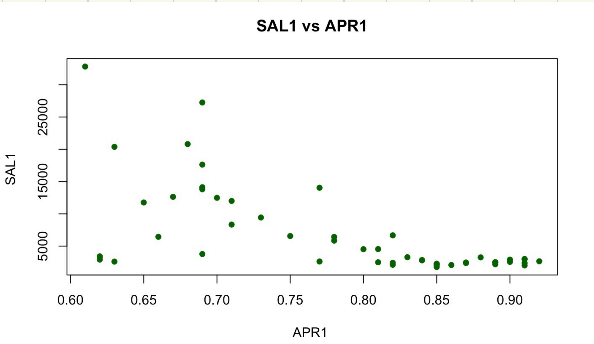
3.31

a.

```
> cat("SAL1  均值 =", round(mean(tuna$sal1)), "| 最小值 =", min(tuna$sal1),
+      "| 最大值 =", max(tuna$sal1), "| 標準差 =", round(sd(tuna$sal1)), "\n")
SAL1  均值 = 6719 | 最小值 = 1762 | 最大值 = 32820 | 標準差 = 6915
> cat("APR1  均值 =", round(mean(tuna$apr1),3), "| 最小值 =", min(tuna$apr1),
+      "| 最大值 =", max(tuna$apr1), "| 標準差 =", round(sd(tuna$apr1),3), "\n")
APR1  均值 = 0.782 | 最小值 = 0.61 | 最大值 = 0.92 | 標準差 = 0.098
```



b.



c.

```
> cat("β2 點估計 (每漲1分錢銷量變化) =", round(b2_hat, 2), "罐\n")
β2 點估計 (每漲1分錢銷量變化) = -434.45 罐
> cat("95% CI for β2 = [", round(ci_b2[1],2), ",", round(ci_b2[2],2), "]\n")
95% CI for β2 = [ -592.29 , -276.6 ]
```

```
========== Part (d) ==========
售價 70 分時預測週銷量   = 10303 罐
售價 70 分時預測週銷量的 90% CI 為  = [ 8625 , 11981 ] 罐

========== Part (e) ==========
平均 PRICE1 = 78.25 分
平均 SAL1              = 6719 罐
彈性估計值             = -5.0598
95% CI for 彈性        = [ -6.8981 , -3.2215 ]
|彈性| > 1?            → 是，高度有彈性 (elastic)

========== Part (f) ==========
H0: 彈性 = -3  vs  H1: 彈性 ≠ -3 (雙尾, 10%)

t 統計量               = -2.2506
臨界值 (雙尾 10%)      = ± 1.6759
p-value                = 0.0288
結論: |t| = 2.25 > 臨界值 1.68 → 拒絕 H0，彈性顯著不等於 -3
```